# Michael Jonathan Susilo

+62 812-2999-771 | mekkjo123@gmail.com | linkedin.com/in/michael-jonathan | github.com/MichaelJonathan110
michael-portofolio-red.vercel.app | Tangerang Selatan, Indonesia

## Summary

Computer Science student and data-engineering practitioner who builds complete, runnable data platforms rather than mockups. Hands-on with ETL/ELT pipelines, dimensional data warehousing (star schema, SCD Type 2), data-quality rule engines and idempotent loading on PostgreSQL. Comfortable owning the full path from raw source files to analytics API and UI. Seeking a Data Engineer internship.

## Education

**BINUS University** Tangerang, Indonesia
*Bachelor of Computer Science — GPA: 3.72 / 4.00* *Sep. 2024 – Present (Expected 2028)*

- Relevant coursework: Data Structures & Algorithms, Database Systems, Data Engineering, Object-Oriented Programming, Statistics.

## Technical Skills

**Languages**: Python, SQL, TypeScript, JavaScript, Java, C
**Data Engineering**: ETL/ELT Pipelines, Data Warehousing, Dimensional Modeling (Star Schema, SCD Type 2), Data Quality Rules, Idempotent Ingestion, Reconciliation, Data Lineage, Batch Processing
**Databases**: PostgreSQL 16, SQLAlchemy 2.0, Alembic, Redis, Schema Design, Query Optimization
**Backend**: FastAPI, REST APIs, Pydantic v2, JWT Authentication, WebSockets
**Frontend**: React 18, TypeScript, Vite, Tailwind CSS, TanStack Query, Progressive Web Apps (PWA)
**Tools & Platforms**: Docker, Docker Compose, Git/GitHub, GitHub Actions (CI), Pentaho Data Integration (Spoon/Carte), Power BI, Tableau, Linux, pytest

## Projects

**RetailFlow — Enterprise Retail Data Warehouse & Analytics Platform** | *Python, FastAPI, PostgreSQL 16, Docker, Next.js, Pentaho, Power BI*

- Architected an end-to-end data warehouse: CSV ingestion → staging → data-quality validation → SCD2 dimensions + fact tables → analytics views → REST API → UI, organized across 5 PostgreSQL schemas.
- Designed a star schema with 5 dimensions (SCD Type 2 on customer) and 2 fact tables; kept facts unjoined to preserve grain correctness (revenue from `fact_order_items`, payments from `fact_payments`) and compared them only as aggregates.
- Built a rules-as-data data-quality engine (12 seeded rules) with rejected-record quarantine (reasons + raw payload) and idempotent ingestion/loads keyed on SHA-256 hashes for exactly-once loads.
- Implemented fact-vs-fact reconciliation with explicit tolerances and recorded variance, plus ETL run and per-stage logging/monitoring.
- Developed a FastAPI backend with 40+ endpoints, OpenAPI docs and a consistent error envelope, and a Next.js 15 UI with 10 pages wired to the live API.
- Wrote a pytest suite including an end-to-end acceptance test: generates a defective fixture, asserts rejects, runs the master pipeline, verifies a second run loads 0 rows, and checks API/fact-table agreement.
- Authored Pentaho (.ktr/.kjb) artifacts and a Power BI semantic model with DAX measures; documented architecture, data dictionary, lineage and deployment.

**Amazon Sales Data Engineering — ETL Pipeline** | *Pentaho PDI, Power BI, SQL*

- Built an ETL pipeline that extracts an Amazon e-commerce sales dataset, cleans/normalizes it and loads a dimensional model (Customer, Product, Location, Time, Status dimensions + Sales fact) using Pentaho Data Integration (Spoon & Carte).
- Delivered an interactive Power BI dashboard surfacing top products/categories, sales trends and regional distribution, backed by a written analysis report.

**Superstore Sales Analytics & Forecasting** | *Tableau, CSV*

- Analyzed 2015–2018 Superstore retail sales in Tableau to surface seasonal trends, top categories/sub-categories, regional distribution and customer-segment performance, including sales forecasting.
- Published an interactive Tableau dashboard and a written analysis translating raw sales data into business insight.

**Trinity — Athlete Performance Platform** | *React 18, TypeScript, FastAPI, PostgreSQL, Redis, Docker*

- Built a full-stack PWA unifying nutrition, recovery, training and activity into one view, with cross-domain intelligence (training readiness, daily performance, per-domain insights).
- Implemented a FastAPI + SQLAlchemy 2.0 backend on PostgreSQL 16 with Redis caching, JWT auth (access + refresh) and bcrypt, containerized with Docker Compose.

**Rally — Social Activity Marketplace** | *FastAPI, SQLAlchemy 2.0, Alembic, PostgreSQL, Redis, WebSockets, React 18, TypeScript*

- Architected a config-driven activity engine (sports, games, outdoor, social, creative) with a layered backend (router → service → repository → model) and server-enforced RBAC across 7 roles.
- Designed idempotent critical flows with audit logging, and provider abstractions for payment/map/email with clearly labelled mock providers.

**LangitNusa — Indonesia Weather & Air-Quality Dashboard** | *React, TypeScript, Vite, Leaflet, PWA*

- Built a near-real-time dashboard integrating 10+ public APIs (Open-Meteo, BMKG, NASA FIRMS, BigDataCloud) with per-panel source labelling (LIVE/FALLBACK/DEMO), timestamps, PWA support and weather-driven dynamic theming.
- Applied data-transparency practices: every panel shows its data source and last-updated time, and modelled $CO_2$ is explicitly labelled as model output rather than a station measurement.

**Personal Portfolio Website** | *TypeScript, React, Vercel*

- Designed and deployed a personal portfolio presenting projects, tech stack and case studies (michael-portofolio-red.vercel.app).